

Problem 11.21

A bank agrees to lend John 10,000 now and X three years later in exchange for a single repayment of 75,000 at the end of 10 years. The bank charges interest at an annual effective rate of 6% for the first 5 years and at a force of interest

$$\delta_t = \frac{1}{t+1}, \quad t \geq 5.$$

Determine X .

Solution

During the first five years, the accumulation function is

$$A(t_1, t_2) = (1.06)^{t_2 - t_1}, \quad 0 \leq t_1 \leq t_2 \leq 5.$$

From time 5 to time 10, the accumulation factor is determined by the force of interest:

$$\begin{aligned} A(5, 10) &= \exp \left(\int_5^{10} \delta_t dt \right) \\ &= \exp \left(\int_5^{10} \frac{1}{t+1} dt \right) \\ &= \exp (\ln(11) - \ln(6)) \\ &= \frac{11}{6}. \end{aligned}$$

Accumulating both loans to time 10 gives

$$10,000(1.06)^5 \left(\frac{11}{6} \right) + X(1.06)^2 \left(\frac{11}{6} \right) = 75,000.$$

The first loan accumulates for five years at 6%, while the second loan, made at time 3, accumulates for two years at 6%. Both then accumulate from time 5 to time 10 by the factor 11/6.

Solving for X ,

$$\begin{aligned} X &= \frac{75,000 \left(\frac{6}{11} \right) - 10,000(1.06)^5}{(1.06)^2} \\ &\approx 24,498.7853. \end{aligned}$$

Therefore,

$$\boxed{X \approx 24,498.79}.$$